

MEET DABGAR

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EDUCATION

National Institute of Technology, Surat (SVNIT)

B.Tech + M.Tech in Artificial Intelligence (Integrated)

2024 – Present

CGPA: 8.53 / 10

RESEARCH EXPERIENCE

Remote Research Intern — CRUNCH Group, Brown University

May 2026 – Nov 2026

Supervised by Prof. Khemraj Shukla & Prof. George Em Karniadakis · Division of Applied Mathematics

- Benchmarking curvature-aware optimizers (SSBFGS, SSBroyden, Natural Gradient, SCIO) on Physics-Informed Neural Networks for compressible flow & 2D Riemann problems — **first systematic optimizer study on neural operator architectures** (DeepONet, FNO), directly assigned by Prof. Karniadakis.
- Investigating κ -EMA stiffness proxy as an **alignment-aware** curvature-conditioned signal for Natural Gradient stabilisation; bridging stiffness detection with principled batch-training dynamics.
- Invited speaker** at CRUNCH Group, Brown University, following independent cold outreach to Prof. Karniadakis.

SCIO — Stiffness-Conditioned Interpolated Optimizer (Independent Research)

Dec 2025 – May 2026

Under Review: IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI), 2026

- Developed SCIO: A stiffness-conditioned optimization framework designed to accelerate the training of Physics-Informed Neural Networks (PINNs) on stiff partial differential equations (PDEs) by adaptively conditioning optimization dynamics.
- Established theoretical foundations including convergence rates and stability guarantees under stiffness conditions; validated performance across **1,044 runs** (9 PDEs, 13 baselines, 10 seeds), demonstrating statistically significant improvements over standard optimizers.

Book Chapter Author — Springer

2025

Hybrid AI Models and Ensemble Learning

- Peer-reviewed chapter on hybrid AI architectures, ensemble techniques, and interpretable intelligent-system design.

INDUSTRY EXPERIENCE

Agentic AI Developer Intern — L&T Technology Services (LTTS)

2026 – Present

On-site · Enterprise AI Automation · One of India's largest technology engineering companies

- Building **production-grade multi-agent pipelines** (LangGraph, LangChain) with human-in-the-loop approval gates and confidence-based routing — directly embodying **AI safety & controllability** principles.
- Deploying interpretable, context-aware automation in live enterprise environments; clean, modular, testable Python.

ML Developer & Mathematics Researcher — Mintzy (FinTech Startup)

2025 · 2 months

Remote · Financial Technology · Early-stage startup

- Rebuilt ML model pipeline from the ground up, improving prediction accuracy from **54% to 69%** — a 15 pp uplift with direct revenue impact; applied quantitative mathematical modelling to redesign core product logic.
- Delivered measurable ML improvements within a two-month engagement working directly with the founding team.

SELECTED PROJECTS

EchoNeRF — Indoor Positioning via WiFi Fingerprinting + Neural Radiance Fields

2025

PowerMind (Adani Airport Hackathon) — 3rd Place

- Zero-hardware indoor navigation for Faraday-cage environments; WiFi + EchoNeRF spatial mapping **outperformed BLE beacon systems**; live demo to judges at the hackathon.

Autonomous Coding Agent — LangGraph · LangChain · HuggingFace Transformers

2024

- Self-correcting LLM agent: graph-based planning, runtime error capture, iterative self-repair; producing multi-file Python projects from natural language with **robust, safety-tested code generation**.

TECHNICAL SKILLS

AI Safety & Interpretability: Mechanistic optimizer analysis, curvature/stiffness probing, human-in-the-loop design, alignment-aware training, uncertainty modelling

Deep Learning: PyTorch, JAX, TensorFlow, HuggingFace Transformers (BERT, GPT, LLaMA, T5), DeepXDE, Physics-Informed Neural Networks, Neural Operators (DeepONet, FNO)

Agentic AI & Backend: LangChain, LangGraph, RAG, Vector Databases, FastAPI, Docker, REST APIs, n8n

Languages & Tools: Python (clean, modular, testable), C, C++, SQL, JavaScript; Git, Linux, Groq API

ACHIEVEMENTS & RECOGNITION

- Invited Speaker**, CRUNCH Group, Brown University — presented independent optimizer research to Prof. George Em Karniadakis after cold outreach.
- IEEE TETCI Under Review** — SCIO optimizer submitted to IEEE Transactions on Emerging Topics in Computational Intelligence, 2026.
- Springer Publication** — peer-reviewed book chapter on Hybrid AI Models and Ensemble Learning, 2025.
- 3rd Place**, PowerMind (Adani Airport Hackathon), 2025.